Multiplying and Dividing Decimals

Count tenths and hundredths like any other things.

3 × 0.4: three groups of 4 tenths



4 tenths + 4 tenths + 4 tenths = 12 tenths.
10 tenths make 1 whole, so 12 tenths = 1.2.

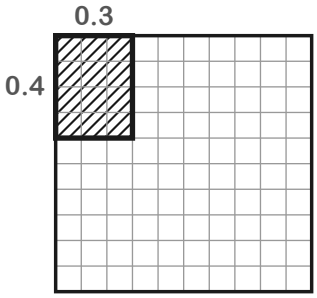
3 × 0.4 = 12 tenths = 1.2

2.4 ÷ 3: share 24 tenths

2.4 is 24 tenths. Share them among 3: each gets 8 tenths.

2.4 ÷ 3 = 8 tenths = 0.8

0.3 × 0.4: a tenth of a tenth is a hundredth



Go 3 tenths across and 4 tenths down.
The rectangle has 3 × 4 = 12 small squares.
Each small square is a hundredth.

0.3 × 0.4 = 0.12

Your turn!

5 × 0.6 = 3

For teachers

Goal: multiply and divide decimals to hundredths with models and place value (5.NBT.7). Say "How many tenths? Count them, then make wholes." No "count the decimal places" rule yet: kids should see why.

Multiplying and Dividing Decimals: Practice

Count the tenths or hundredths. 10 tenths make 1.

Part 1 Count the tenths How many tenths in all? Then write the decimal.

1) $4 \times 0.3 =$ 12 tenths = 1.2

2) $6 \times 0.7 =$ 42 tenths = 4.2

3) $2 \times 0.9 =$ 18 tenths = 1.8

4) $5 \times 0.8 =$ 40 tenths = 4

5) $7 \times 0.4 =$ 28 tenths = 2.8

6) $3 \times 0.6 =$ 18 tenths = 1.8

7) $8 \times 0.5 =$ 40 tenths = 4

8) $9 \times 0.2 =$ 18 tenths = 1.8

9) $6 \times 0.4 =$ 24 tenths = 2.4

10) $7 \times 0.9 =$ 63 tenths = 6.3

Multiplying and Dividing Decimals: Practice (continued)

Part 2 Count the hundredths How many hundredths in all? Then write the decimal.

11) $3 \times 0.25 =$ [75] hundredths = [0.75]

12) $4 \times 0.18 =$ [72] hundredths = [0.72]

13) $2 \times 0.46 =$ [92] hundredths = [0.92]

14) $5 \times 0.13 =$ [65] hundredths = [0.65]

15) $6 \times 0.15 =$ [90] hundredths = [0.9]

16) $3 \times 0.32 =$ [96] hundredths = [0.96]

17) $7 \times 0.12 =$ [84] hundredths = [0.84]

18) $4 \times 0.35 =$ [140] hundredths = [1.4]

Part 3 Split the decimal Ones first, then tenths.

19) $6 \times 1 + 6 \times 0.4 =$ [6] + [2.4] = [8.4]

20) $3 \times 2 + 3 \times 0.7 =$ [6] + [2.1] = [8.1]

21) $4 \times 3 + 4 \times 0.6 =$ [12] + [2.4] = [14.4]

22) $7 \times 1 + 7 \times 0.8 =$ [7] + [5.6] = [12.6]

Multiplying and Dividing Decimals: Practice (continued)

Part 3 Split the decimal (continued)

23) $5 \times 4 + 5 \times 0.3 = \boxed{20} + \boxed{1.5} = \boxed{21.5}$

24) $8 \times 2 + 8 \times 0.5 = \boxed{16} + \boxed{4} = \boxed{20}$

Part 4 Multiply No help this time.

25) $3 \times 0.7 = \boxed{2.1}$

26) $4 \times 2.3 = \boxed{9.2}$

27) $6 \times 0.15 = \boxed{0.9}$

28) $5 \times 1.9 = \boxed{9.5}$

29) $2 \times 3.45 = \boxed{6.9}$

30) $8 \times 0.6 = \boxed{4.8}$

31) $7 \times 2.4 = \boxed{16.8}$

32) $9 \times 0.08 = \boxed{0.72}$

33) $3 \times 4.25 = \boxed{12.75}$

Multiplying and Dividing Decimals: Practice (continued)

Part 4 Multiply (continued)

34) $6 \times 1.7 =$ 10.2

35) $4 \times 0.35 =$ 1.4

36) $5 \times 6.2 =$ 31

37) $8 \times 1.25 =$ 10

38) $2 \times 7.8 =$ 15.6

39) $9 \times 0.4 =$ 3.6

Part 5 Share the tenths How many tenths? Share them.

40) 2.4 ÷ 3: 24 tenths ÷ 3 = 8 tenths = 0.8

41) 3.5 ÷ 5: 35 tenths ÷ 5 = 7 tenths = 0.7

42) 4.8 ÷ 6: 48 tenths ÷ 6 = 8 tenths = 0.8

43) 1.6 ÷ 4: 16 tenths ÷ 4 = 4 tenths = 0.4

44) 6.3 ÷ 7: 63 tenths ÷ 7 = 9 tenths = 0.9

45) 5.6 ÷ 8: 56 tenths ÷ 8 = 7 tenths = 0.7

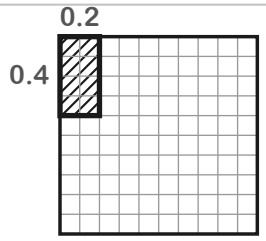
Multiplying and Dividing Decimals: Practice (continued)

Part 6 Divide No help this time.

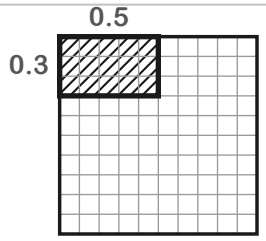
46) $1.8 \div 3 = 0.6$	47) $4.5 \div 5 = 0.9$	48) $0.84 \div 4 = 0.21$
49) $7.2 \div 8 = 0.9$	50) $3.6 \div 6 = 0.6$	51) $0.96 \div 3 = 0.32$
52) $8.1 \div 9 = 0.9$	53) $2.5 \div 5 = 0.5$	54) $1.48 \div 2 = 0.74$
55) $6.4 \div 4 = 1.6$	56) $0.75 \div 3 = 0.25$	57) $9.6 \div 8 = 1.2$

Multiplying and Dividing Decimals: Practice (continued)

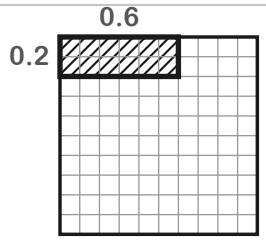
Part 7 Tenths times tenths Shade the rectangle, then count the small squares.

58) 

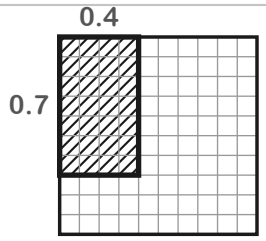
$0.2 \times 0.4 =$ 0.08

59) 

$0.5 \times 0.3 =$ 0.15

60) 

$0.6 \times 0.2 =$ 0.12

61) 

$0.4 \times 0.7 =$ 0.28

Part 8 Tenths times tenths No help this time.

62) $0.6 \times 0.7 =$ 0.42

63) $0.8 \times 0.3 =$ 0.24

64) $0.9 \times 0.5 =$ 0.45

65) $0.3 \times 0.3 =$ 0.09

66) $0.7 \times 0.2 =$ 0.14

67) $0.4 \times 0.9 =$ 0.36

Multiplying and Dividing Decimals: Practice (continued)

Part 8 Tenths times tenths (continued)

68) $0.5 \times 0.6 = 0.3$

69) $0.8 \times 0.8 = 0.64$

70) $0.2 \times 0.9 = 0.18$

Part 9 Multiply

71) $5 \times 1.8 = 9$

72) $9 \times 1.3 = 11.7$

73) $8 \times 5.8 = 46.4$

74) $9 \times 0.5 = 4.5$

75) $3 \times 4.3 = 12.9$

76) $8 \times 6.3 = 50.4$

77) $6 \times 1.2 = 7.2$

78) $9 \times 0.2 = 1.8$

79) $8 \times 0.9 = 7.2$

Multiplying and Dividing Decimals: Practice (continued)

Part 9 Multiply (continued)

80) $9 \times 3.9 = 35.1$

81) $5 \times 3.3 = 16.5$

82) $6 \times 6.7 = 40.2$

83) $9 \times 0.7 = 6.3$

84) $8 \times 1.3 = 10.4$

85) $9 \times 2.1 = 18.9$

86) $7 \times 0.7 = 4.9$

87) $9 \times 6.1 = 54.9$

88) $5 \times 1.7 = 8.5$

Part 10 Divide

89) $11.9 \div 7 = 1.7$

90) $4.2 \div 7 = 0.6$

91) $2.1 \div 3 = 0.7$

Multiplying and Dividing Decimals: Practice (continued)

Part 10 Divide (continued)

92) $3.8 \div 2 =$ 1.9

93) $15 \div 6 =$ 2.5

94) $4 \div 5 =$ 0.8

95) $7.2 \div 4 =$ 1.8

96) $25.2 \div 9 =$ 2.8

97) $9.6 \div 4 =$ 2.4

98) $5.1 \div 3 =$ 1.7

99) $1.2 \div 2 =$ 0.6

100) $5.6 \div 8 =$ 0.7

101) $8.7 \div 3 =$ 2.9

102) $1.2 \div 3 =$ 0.4

103) $5.5 \div 5 =$ 1.1

Multiplying and Dividing Decimals: Practice (continued)

Part 10 Divide (continued)

104) $10.2 \div 6 =$ 1.7

105) $1.2 \div 6 =$ 0.2

106) $4.8 \div 4 =$ 1.2

Part 11 Tenths times tenths

107) $0.1 \times 0.1 =$ 0.01

108) $0.1 \times 0.2 =$ 0.02

109) $0.1 \times 0.3 =$ 0.03

110) $0.1 \times 0.4 =$ 0.04

111) $0.1 \times 0.5 =$ 0.05

112) $0.1 \times 0.6 =$ 0.06

113) $0.1 \times 0.7 =$ 0.07

114) $0.1 \times 0.8 =$ 0.08

115) $0.1 \times 0.9 =$ 0.09

Multiplying and Dividing Decimals: Practice (continued)

Part 11 Tenths times tenths (continued)

116) $0.2 \times 0.1 =$ **0.02**

117) $0.2 \times 0.2 =$ **0.04**

118) $0.2 \times 0.3 =$ **0.06**

Part 12 Remember? Problems from earlier packets.

119) $858 \div 26 =$ **33**

120) $215 \times 34 =$ **7,310**

121) $4.6 + 0.85 =$ **5.45**

122) 3.74 to the nearest tenth = **3.7**

123) 0.32 **<** 0.4

124) $2 + 0.4 + 0.005 =$ **2.405**

Part 13 Word problems Count the tenths and hundredths.

125) A pencil costs $0.35. How much do 6 pencils cost?

Number sentence **6 × 0.35 = 2.10**

Answer **$2.10**

Draw or show your work

126) A ribbon is 4.8 m long. Ava cuts it into 4 equal pieces. How long is each piece?

Number sentence **4.8 ÷ 4 = 1.2**

Answer **1.2** m

Draw or show your work

127) A plant grows 1.5 cm each week. How much does it grow in 7 weeks?

Number sentence **7 × 1.5 = 10.5**

Answer **10.5** cm

Draw or show your work

Multiplying and Dividing Decimals: Practice (continued)

Part 13 Word problems (continued)

128) Three friends share $7.50 equally. How much does each friend get?

Number sentence **7.50 ÷ 3 = 2.50**

Answer **$2.50**

Draw or show your work

129) Challenge: A garden is 0.6 km long and 0.8 km wide. What is its area?

Number sentence **0.6 × 0.8 = 0.48**

Answer **0.48** sq km

Draw or show your work

Check-In: Multiplying and Dividing Decimals

Show what you know. No help this time.

Mastery: 9 of 11 (80%)

Part 1 Count the tenths

1) $3 \times 0.9 =$ 27 tenths $=$ 2.7

2) $6 \times 0.6 =$ 36 tenths $=$ 3.6

Part 2 Multiply

3) $2 \times 4.15 =$ 8.3

4) $5 \times 2.8 =$ 14

5) $3 \times 1.6 =$ 4.8

Part 3 Divide

6) $1.2 \div 4 =$ 0.3

7) $5.4 \div 6 =$ 0.9

8) $0.63 \div 3 =$ 0.21

Part 4 Tenths times tenths

9) $0.3 \times 0.8 =$ 0.24

10) $0.6 \times 0.5 =$ 0.3

11) $0.2 \times 0.6 =$ 0.12